

Calculated Fields, Parameters & LOD

Expressions Assignment

Theoretical Questions:

Question 1: What is a Calculated Field and why is it important in business dashboards?

Answer:

A Calculated Field is a **new field/column that you create yourself in Tableau** using a formula it does not exist in your original dataset but is calculated on the fly using existing fields.

Think of it like an **Excel formula** you use existing columns to calculate something new.

Example:

Profit Margin % = $\text{SUM}(\text{Profit}) / \text{SUM}(\text{Sales}) * 100$

Why is it Important in Business Dashboards?

Reason 1 — Raw data is never enough:

- Datasets give you Sales and Profit separately
- But the CEO wants **Profit Margin %** — which requires calculation

Reason 2 — Creates business specific KPIs:

- Every business has unique metrics — Profit Margin, Return Rate, Growth %
- These **don't come ready made** in any dataset

Reason 3 — Makes dashboards dynamic and intelligent:

- Instead of static numbers, calculated fields respond to **filters and interactions**
- When you filter by Region — the Profit Margin % **automatically recalculates** for that region

Reason 4 — Saves time and reduces errors:

- Without calculated fields you'd need to **pre-process data in Excel** every time
- In Tableau you define the formula once → it **applies everywhere automatically**

Question 2: Why can't parameters change visuals directly in Tableau?

Answer:

Parameters **alone cannot change visuals directly** in Tableau because a parameter is just a **variable that stores a value** it has no power to do anything on its own until you **connect it to something** like a Calculated Field, Filter, or Reference Line.

Think of a parameter like a **remote control without batteries** it exists but does nothing until you connect it to power.

Question 3 : When should EXCLUDE LOD be used?

Answer:

LOD stands for **Level of Detail** it lets you control **at what level Tableau calculates a value**, independent of what is shown in the view.

EXCLUDE removes a dimension from the calculation that is currently present in the view forcing tableau to calculate at a higher/broader level when still showing detail in the view.

Question 4 : Difference between FIXED and INCLUDE LOD expressions? Practical

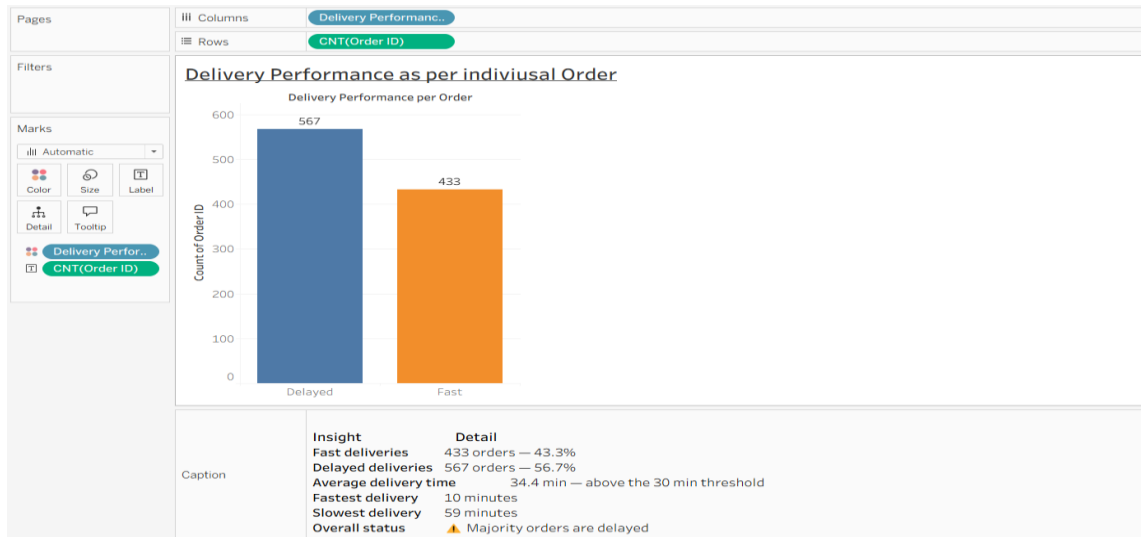
Answer:

LOD Types & there explanation	FIXED	INCLUDE
What it does	Calculates at YOUR specified level ignores the view completely	Adds extra dimension TO the current view level
Affected by view?	No rigid, stays fixed	Yes adds onto whatever is in view
Affected by filters?	No (ignores dimension filters)	Yes
Control level	You decide exactly	Depends on view + what you add

Practical Questions :

Task 1: Delivery Performance Classification (Calculated Field) Business Need Operations wants to track delivery efficiency. Task Classify orders as Fast or Delayed. Condition $\text{Delivery} \leq 30 \text{ min} \rightarrow \text{Fast}$ Else $\rightarrow \text{Delayed}$

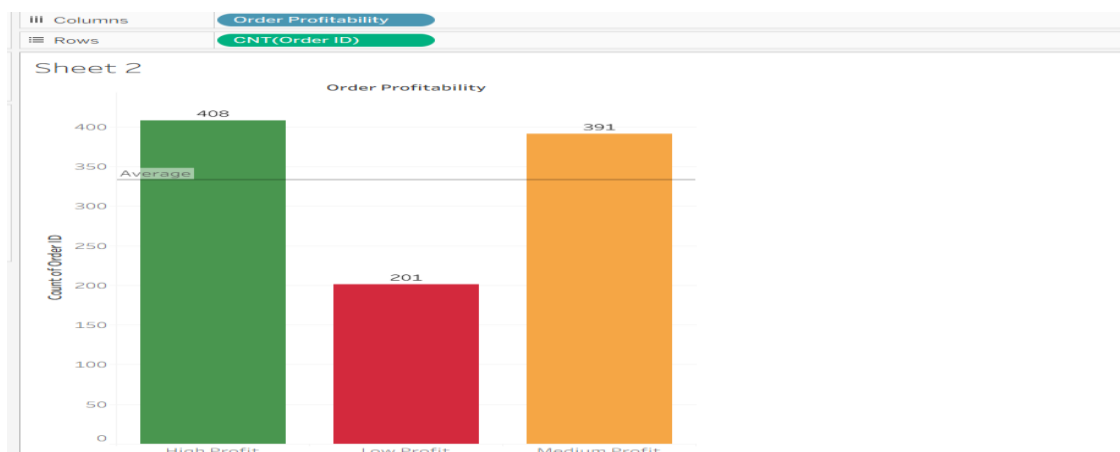
Answer:



Sheet1	#	Abc	#	#	#	Abc
Unit Price	Delivery Time Min	Payment Mode	Sales	Cost	Profit	Delivery Performance per
173	20	Cash	346	238.076	107.924	Fast
128	37	UPI	512	312.717	199.283	Delayed
167	55	Cash	334	215.354	118.646	Delayed
135	41	UPI	135	94.237	40.763	Delayed
106	14	UPI	212	177.432	34.568	Fast
29	47	UPI	29	18.424	10.576	Delayed

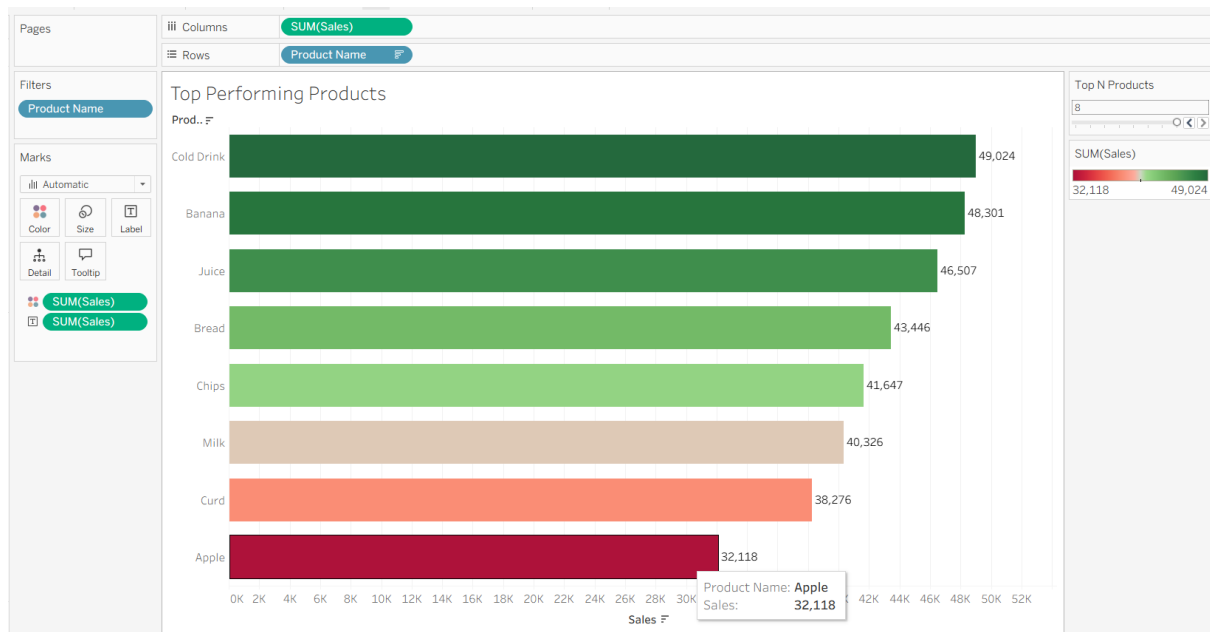
Task 2: Loss-Making Orders Identification Blinkit Orders Assignment Questions Pwskills Business Need High sales do not always mean high profit. Task Identify orders resulting in loss.

Answer:



Task 3: Top N Products Analysis (Parameter) Business Need The product team wants to focus on top-performing products. Task Create a Top N Products dynamic analysis.

Answer:



The **Top N Parameter** makes the dashboard dynamic letting the product team instantly switch between Top 3, Top 5 or all 8 products revealing that **Cold Drink leads in Sales** but **Banana is the true profit champion** that deserves maximum business attention.

Task 4: City-Level Benchmarking (FIXED LOD) Business Need Compare city performance independent of category filters. Task Calculate average sales per city using FIXED LOD.

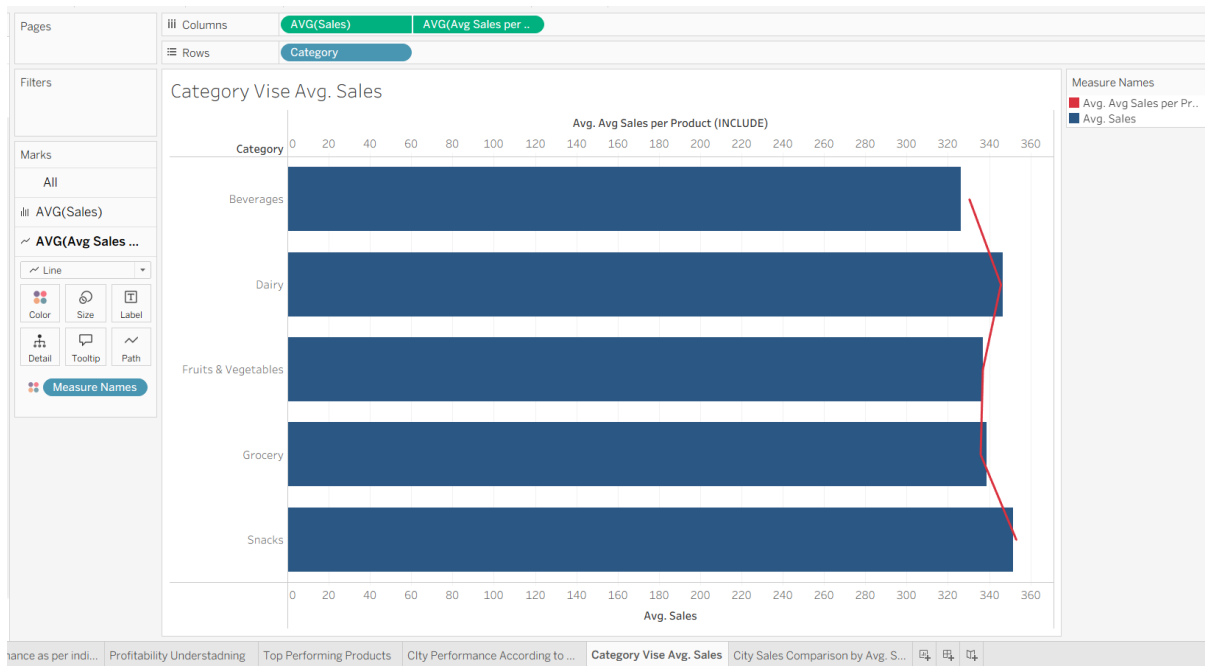
Answer:



Task 5: Customer/Product Detail Impact (INCLUDE LOD) Business Need The sales team wants average sales considering product-level detail, even when product is not shown. Task Create a calculation that finds the average sales at the product level, but allows the result to be used at a higher level (such as Category or Region). Use INCLUDE LOD.

Answer:

Category Vise comparison



City Comparison

